MATHEMATICS 2210

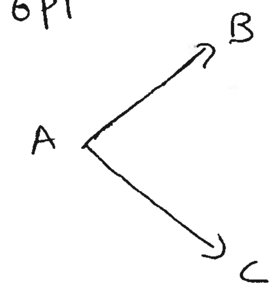
Quiz 2, Fall semester 2012

Name: KEY

1. Find the area of the triangle with vertices $(0,3,2)$, $(1,2,0)$ and $(3,-1,1)$.

A B C

6pt



$$\vec{AB} = (1,-1,-2)$$

$$\vec{AC} = (3,-4,-1)$$

$$\vec{AB}\times\vec{AC} = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ 1 & -1 & -2 \\ 3 & -4 & -1 \end{vmatrix} = \vec{i}\begin{vmatrix} -1 & -2 \\ -4 & -1 \end{vmatrix} - \vec{j}\begin{vmatrix} 1 & -2 \\ 3 & -1 \end{vmatrix} + \vec{k}\begin{vmatrix} 1 & -1 \\ 3 & -4 \end{vmatrix}$$

$$= (-7, -5, -1)$$

$$\text{Area} = \frac{\|\vec{AB}\times\vec{AC}\|}{2} = \frac{\sqrt{49+25+1}}{2} = \frac{\sqrt{75}}{2} = \frac{5}{2}\sqrt{3}$$

2. Find the volume of the parallelepiped with edges $(3,-1,1)$, $(2,0,3)$ and $(1,-4,2)$.

$$\begin{vmatrix} 3 & -1 & 1 \\ 2 & 0 & 3 \\ 1 & -4 & 2 \end{vmatrix} = 3\begin{vmatrix} 0 & 3 \\ -4 & 2 \end{vmatrix} + 1\begin{vmatrix} 2 & 3 \\ 1 & 2 \end{vmatrix} + 1\begin{vmatrix} 2 & 0 \\ 1 & -4 \end{vmatrix}$$

$$= 3(0+12) + 1(4-3) + 1(-8+0) =$$

$$= 36 + 1 - 8 = 29$$